S S JHOTHEESHWAR

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Education

Chennai Institute Of Technology **2024-2028**
B.E. Electronics Engineering (VLSI Design and Technology) **CGPA : 8.4**

Internship Experience

Switch Mobility (Ashok Leyland) Internship, Ennore **MAY - JUN 2025**

• Studied **CAN** communication and the interaction between **ECUs**, understanding how vehicle subsystems exchange data for control, monitoring, and diagnostics.

• Acquired practical knowledge of automotive sensors such as **torque sensors**, **steering angle sensors**, and inductive proximity sensors used in EV systems.

India Meteorological Department (IMD) - Regional Meteorological Centre, Chennai **MAY - JUN 2026**

• Acquired practical expertise in **Automatic Weather Stations (AWS)**, **Automatic Rain Gauges (ARG)**, meteorological instrumentation, data loggers, **GSM**-based telemetry, and **WMO-compliant** weather observation practices.

• Developed a Python-based AWS–ARG Quality Control & Health Monitoring platform, **currently used daily by the Regional Meteorological Centre (RMC), Chennai** for automated quality control, **sensor validation**, fault analysis, and operational reporting.

Projects

Vision-x - Real-Time Vision AI Object Detection System on Renesas RA8P1 using µT-Kernel 3.0 Github Link **JUN 2026**

• Developed a real-time Vision AI system on the **Renesas RA8P1** using **µT-Kernel 3.0** (TRON RTOS), designing a multithreaded architecture for concurrent camera capture, **NPU-based AI inference**, and LCD rendering with deterministic real-time performance.

• Integrated and optimized an INT8 YOLOX object detection model on the Arm Ethos-U55 NPU, implementing CPU–NPU shared-memory synchronization.

LATTEYES - Multi-Purpose Edge-AI Surveillance System using Lattice CertusPro-NX FPGA Github Link **JAN 2026**

• Designed and implemented a 4-mode **Edge-AI surveillance system** (Intruder, Wildlife, Industrial Fire, Border/Drone) on Lattice CertusPro-NX FPGA, achieving 92% object detection accuracy

• Implemented MobileNetV2-SSD on an FPGA-based **CNN accelerator**, enabling efficient real-time object detection at the edge.

• Deployed HM0360 (640×480), dual-PIR sensors, **AXI Interface**, enabling real-time bounding box rendering and live annotated video streaming (cloud-independent).

SafeLink - Secure Real Time Sensor Communication System Using SPN Encryption (ZedBoard FPGA) Github Link **FEB 2026**

• Established a hardware-based secure sensor communication system on **Zynq-7000 FPGA** using Verilog RTL, achieving deterministic real-time operation without processor involvement.

• Programmed a 8-bit SPN encryption engine **(XOR key mixing, 4-bit S-box, permutation layer, checksum)** transmitting encrypted data via UART.

PiAssistant - Offline Hindi Voice Assistant using Raspberry Pi 4 Github Link **FEB 2026**

• Developed an offline Hindi voice assistant on **Raspberry Pi 4** integrating local speech recognition, **LLM inference**, and **text-to-speech** for privacy-preserving voice interaction.

• Integrated Whisper, **Ollama (Gemma)**, and Piper into a real-time voice pipeline optimized for low-latency, on-device AI execution without cloud services.

Skills

- **Languages and Libraries/Frameworks:** C, Java, Verilog, Python.
- **Microcontrollers/Microprocessors:** ESP32, Aries V2, Raspberry Pi 4 , Renesas RA8P1, RA6E2.
- **FPGAs:** Lattice Certus Pro-NX, Tang nano-9K, ZedBoard - Zynq-7000, Kria KV260.
- **Communication/Network Protocols:** UART, SPI, I2C, I2S, CAN, USB, BLE, WiFi, JTAG, AXI.
- **AI/ML:** PyTorch, Numpy, TensorFlow, Keras, OpenCV, ONNX, Matplotlib, imgaug.
- **Software & Tools:** Synopsys, Cadence, Xilinx Vivado, e2 studio, Lattice Radiant, GitHub, VSCode, Lam Research Semulator 3D, LTspice.

Achievements

- VLSID 2026 (Design Contest) -**Winner** for an FPGA-based Edge-AI Surveillance System at the VLSID 2026 International Conference.
- Tron Programming Contest 2026 (Finalist) - for developing a real-time Vision AI application on the Renesas RA8P1 using µT-Kernel 3.0
- IIT Bombay - Safetronics Finalist (DEC 2025)

Certifications

- **NPTEL :** Introduction to Industry(4.0), System Design through Verilog, Sensors and Actuators
- **EduSkills :** MICROCHIP EMBEDDED SYSTEM DEVELOPER VIRTUAL INTERNSHIP

Coding Platforms: Leetcode**:** Solved 191 **Rating:** 1,281 | Codechef**:** Solved 51 **Rating:** 785 | Skillrack**:** Solved 138